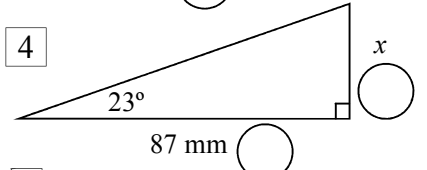
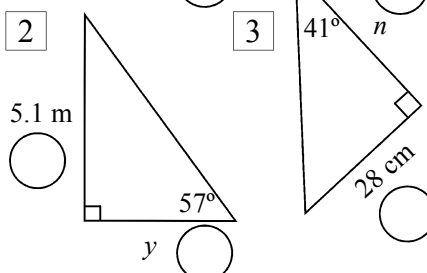
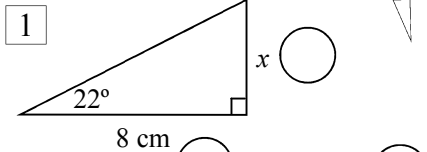


Trigonometry - Finding a Side (tan) (Sheet 1)

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Label these sides 'O', and 'A'.



Calculate these, answer to 1 d.p.

5 $14 \tan 53^\circ =$

Calc: 14TAN53=

6 $9 \tan 35^\circ =$

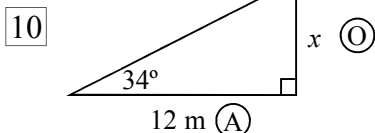
7 $43.8 \tan 32^\circ =$

8 $\frac{77}{\tan 63^\circ} =$

9 $\frac{120}{\tan 46^\circ} =$

Calc: 77÷TAN63=

Calculate the side length. Answering to 1 decimal place.

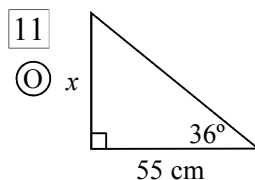


$$\tan 34 = \frac{\text{O}}{\text{A}}$$

($\times 12$) ($\times 12$)

$$x = \text{ } \tan \text{ }$$

$$x = \text{ } \text{ m } \quad 1 \text{ d.p.}$$

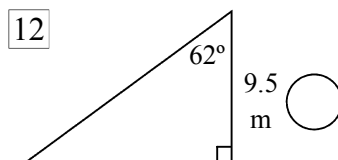


$$\tan 36 = \frac{\text{O}}{\text{A}}$$

($\times 55$) ($\times 55$)

$$x = \text{ } \tan \text{ }$$

$$x = \text{ } \text{ cm } \quad 1 \text{ d.p.}$$



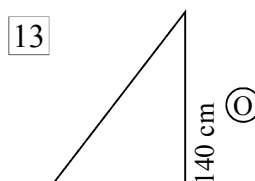
$$\tan 62 = \frac{\text{O}}{\text{A}}$$

($\times 9.5$) ($\times 9.5$)

$$k = \text{ } \tan \text{ }$$

$$k = \text{ } \text{ m } \quad 1 \text{ d.p.}$$

Now 'x' (the term) is on the bottom



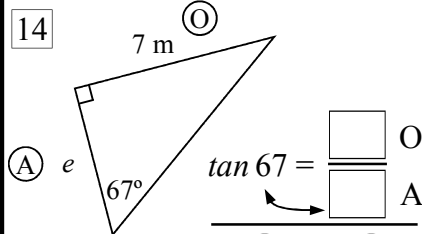
$$\tan 54 = \frac{\text{O}}{\text{A}}$$

(swap) (swap)

$$x = \frac{\text{O}}{\text{A}}$$

$$x = \text{ } \text{ cm }$$

1 d.p.



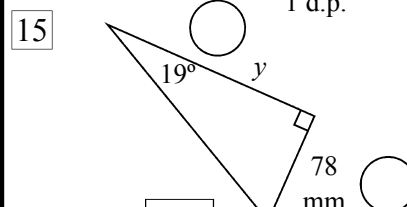
$$\tan 67 = \frac{\text{O}}{\text{A}}$$

(swap) (swap)

$$e = \frac{\text{O}}{\text{A}}$$

$$e = \text{ } \text{ m }$$

1 d.p.



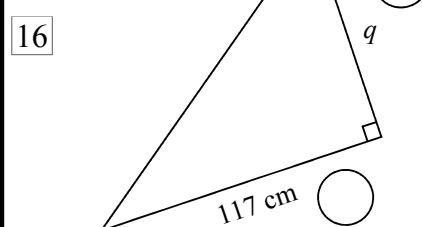
$$\tan 19 = \frac{\text{O}}{\text{A}}$$

(swap) (swap)

$$y = \frac{\text{O}}{\text{A}}$$

$$y = \text{ } \text{ mm }$$

1 d.p.



$$\tan 73 = \frac{\text{O}}{\text{A}}$$

(swap) (swap)

$$q = \frac{\text{O}}{\text{A}}$$

$$q = \text{ } \text{ cm }$$

1 d.p.

Trigonometry - Finding a Side (tan) (Sheet 2)

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Find the side length (1 d.p.).
All are tan, but are a mix of 'x on top' and 'x on the bottom'.

1

38° 13 cm x

$\tan \square = \frac{\square}{\square}$

$\times 13 \quad \times 13$

$x = \square \tan \square$

$x = \square \text{ cm} \quad 1 \text{ d.p.}$

2

6.3 m 26° d

$\tan \square = \frac{\square}{\square}$

swap swap

$d = \frac{\square}{\tan \square}$

$d = \square$

Remember the units

3

30° 38 cm n

$\tan \square = \frac{\square}{\square}$

$n = \square \tan \square$

$n = \square$

4

9 m 55° x

$\square$

$\square$

$\square$

5

72° 32 m c

$\square$

$\square$

$\square$

6

38° y 25 cm

$\square$

$\square$

$\square$

7

53 m 81° x

$\square$

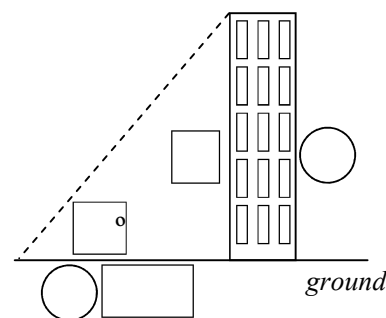
$\square$

$\square$

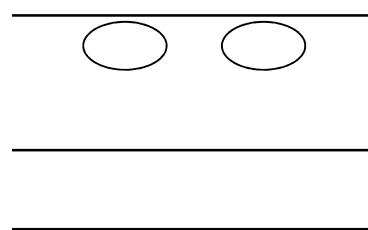
Answer these using your tan trig skills. Find the length to 1 decimal place.

Jess looks at a building that is 72 m away. She looks up at an angle of 54°

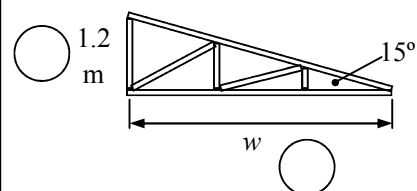
8 Write 'x' as the height of the building, then write the angle and the distance in the boxes on the diagram.



9 Label the distances 'O' and 'A' then calculate the building's height.



A roof truss is angled at 15° and has a height of 1.2 m. The width of the roof 'w' is shown on the diagram.



10 Find 'w', the width of the roof.

